

Three Gates to a Quark Generation

An Exact Criterion for Which $SU(4)$ Representations Contain the Standard Model, and Why the Gauge Group Is Critical at Rank Three

Part II — the $\pm\frac{1}{2}$ cell, a Levi branching, and the orbifold chiral projection

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July 18, 2026

DOI: [10.5281/zenodo.21432628](https://doi.org/10.5281/zenodo.21432628)

Abstract

The companion paper embedded a full Standard-Model quark generation in the dimension-60 representation $(3, \mathbf{60})$ of $SU(4)$ on $T^2/\mathbb{Z}_2$ and showed, by an exhaustive scan, that $(3, \mathbf{60})$ is the *smallest* representation able to do so. That minimality begs the general question: *which* $SU(4)$ representations can hold a quark generation, and why only those? We answer it exactly. An irreducible representation with Dynkin labels (a, b, c) contains the Standard-Model quark cell $\{Q(\mathbf{2}, \frac{1}{6}), u(\mathbf{1}, \frac{2}{3}), d(\mathbf{1}, -\frac{1}{3})\}$ in its $T^2/\mathbb{Z}_2$ chiral zero modes if and only if

$$(a + 2b + 3c) \text{ odd} \quad \wedge \quad b \geq 1 \quad \wedge \quad a + b + c \geq 3.$$

The three clauses are, in order, an *arithmetic* condition (the $\mathbb{Z}_4$ centre charge), a *geometric* one (the middle node of $SU(4)$ excited), and a *size* one (enough room to close the cell). We derive the criterion by deleting the middle node of the $SU(4)$ Dynkin diagram — exposing a Levi subgroup $SU(2)_L \times SU(2)_R \times U(1)$ — and projecting the resulting Clebsch–Gordan tower onto the orbifold’s chiral zero modes; the projection collapses the tower to a closed-form zero-mode count $N = (b + 1)(a + c + 1)/2$. We are explicit about what is new and what is inherited: the centre gate is classical, the branching is Littlewood–Richardson, and the midpoint property (the doublet’s $\frac{1}{6}$ is the mean of the singlets’ $\frac{2}{3}$ and $-\frac{1}{3}$) reflects the Pati–Salam hypercharge relation; what is ours is the chiral projection that yields the closed count, the packaged criterion, and the reading of the cell as the invariant object. Finally we place $SU(4)$ in a family: the cell imposes three charge constraints on a hypercharge functional with $\text{rank}(G) - 1$ free parameters, so it is a *rigid* obstruction exactly at rank = 3 — $SU(4)$ is the marginal case, and for higher rank a generic representation admits. The same counting explains, as a consumed-degree-of-freedom effect, why a pinned hypercharge in higher-rank models leaves quarks with the wrong (integer) charges and forces them onto a brane.

1 From one representation to all of them

This paper continues a story its companion began. Part I [1] took up a question open since 2003, when Scrucca, Serone, Silvestrini and Wulzer observed that on an orbifold the vanishing of localized tadpoles and the cancellation of anomalies are *independent* requirements, and named the missing piece — an anomaly-free spectrum with a globally vanishing tadpole — as an open problem [2]. Part I met it in the six-dimensional two-Higgs-doublet $SU(4)$ model of Akamatsu,

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Hirose, Maru and Nago [3]: it embedded a colored quark block in the dimension-60 representation $(3, \mathbf{60})$, exhibited a 45-multiplet completion cancelling anomalies and tadpole together, and proved an *Existence theorem* — on that fermion library the tadpole never obstructs an anomaly-free completion — turning a computational witness into a structural guarantee. It closed with a proposal: wherever a model-building result reads “we found one,” ask whether the find was in fact a law. This paper takes Part I at its word. For in passing, and without explaining why, Part I had recorded a second find — an exhaustive scan showing $(3, \mathbf{60})$ to be the *smallest* $SU(4)$ representation able to host a full quark generation. Why that one, and why so few? We turn the scanned fact into an exact law.

That question is older than the model, and its history is one of boxes packed ever tighter. A grand-unified or gauge–Higgs construction begins by choosing a representation to carry the matter, and the choice is severely constrained: the representation must contain, among its states, exactly the quantum numbers of one generation. Georgi and Glashow fit a generation into the $\mathbf{\bar{5}} \oplus \mathbf{10}$ of $SU(5)$ [5] — in two pieces, and with no seat for a right-handed neutrino. A year later the $\mathbf{16}$ spinor of $SO(10)$ closed the box: fifteen Standard-Model fermions, and a sixteenth place the spinor simply *has*, all but forcing one to seat a ν_R there [6, 7] — the group knowing, as it were, before we did. The exceptional $\mathbf{27}$ of E_6 followed, volunteering a generation plus exotics [8], and Georgi asked outright for all three families in one irreducible representation [9, 11], a dream that founders on a hall of mirror fermions. Through all of it Slansky’s tables were the field’s phone-book for deciding what a representation contains [10]; the recent census of Herms and Ruhdorfer turns the tallies into a statistic — what fraction of anomaly-free matter is unifiable [12]. But the decision stayed, representation by representation, a lookup rather than a law.

A minimum is an invitation. If one representation is smallest, there is a landscape behind it — and Part I mapped a single point of it. What distinguishes the representations that succeed from those that fail? Is “ $(3, \mathbf{60})$ ” an accident of dimension, or the first member of a family with a clean rule? This paper gives the rule, exactly, and reads off from it why the family is so sparse, why the minimum falls where it does, and why the gauge group is $SU(4)$ and not something larger. Where Part I validated a witness, Part II charts the space around it; together they run from one consistent model to an understanding of the whole matter landscape it lives in.

The organizing object throughout is what we will call the *Standard-Model cell*: the color-triplet quark content of one generation, stripped to its electroweak skeleton,

$$Q(\mathbf{2}, \tfrac{1}{6}), \quad u(\mathbf{1}, \tfrac{2}{3}), \quad d(\mathbf{1}, -\tfrac{1}{3}).$$

The up and down singlets are the doublet displaced in hypercharge by $\pm \frac{1}{2}$ — the right-handed isospin T_R^3 of the Pati–Salam relation we meet in §2. The operative fact is a small piece of arithmetic: the doublet’s $\frac{1}{6}$ is the *mean* of the singlets’ charges, $\frac{1}{6} = \frac{1}{2}(\frac{2}{3} + (-\frac{1}{3}))$, so the doublet sits at the exact hypercharge midpoint of its two singlets. That midpoint is what makes the cell a rigid geometric object in the weight lattice, and it is where our criterion begins.

Scope and relation to prior work. We are careful to separate what is new from what is inherited, and most of the ingredients are inherited. The centre-charge gate — that admitting the Standard Model’s fractional hypercharges forces a fixed congruence class of the gauge-group centre — is classical, stated for the Standard Model’s own $\mathbb{Z}_6$ by Hucks and by Saller and revisited by Tong and by Alonso, Dimakou and West [13, 14, 15, 16]. The branching we use is a maximal-parabolic Levi restriction, whose multiplicities are Littlewood–Richardson coefficients [19, 20]. The midpoint relation is the Pati–Salam hypercharge identity, and the observation that a hypercharge functional embedded in a simple group has a fixed number of free parameters is the same counting that underlies the classic “rank ≥ 4 ” lower bound on unified groups; a closely related Cartan-functional analysis of gauge-Higgs models, though of the gauge-boson sector rather than the matter cell, is due to Aranda and Wudka [17]. We claim no new group theory. What is ours is threefold: (i) the *exact three-gate criterion* for $SU(4)$ on $T^2/\mathbb{Z}_2$, packaged as a single closed statement (§5); (ii) the *orbifold chiral projection* that collapses the Levi Clebsch–Gordan tower

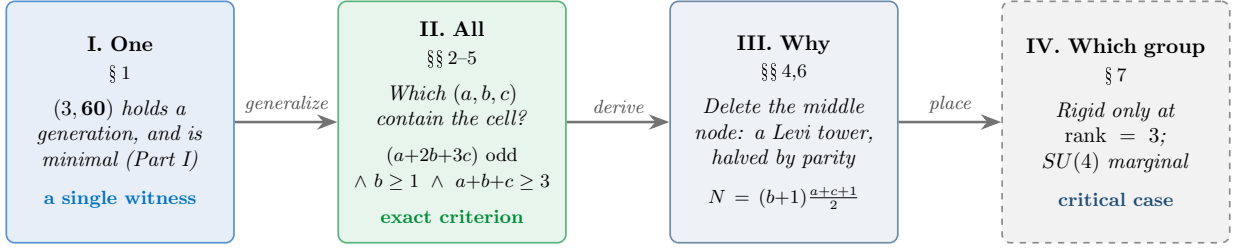


Figure 1: The route of this paper. From the single minimal witness of Part I to an exact criterion for all representations; the criterion is derived by a Levi branching and a chiral projection, and $SU(4)$ is then placed as the unique marginal rank at which the cell is a rigid obstruction.

to the closed-form zero-mode count $N = (b+1)(a+c+1)/2$ (§4); and (iii) the reading of the $\pm\frac{1}{2}$ cell as the invariant, together with the placement of $SU(4)$ at the critical rank (§7) — the last an interpretive lens on classical ingredients, not a new theorem. Figure 1 is the route.

2 The Standard-Model cell as a lattice problem

What does it mean, exactly, for a representation to *contain* the Standard Model — and can we decide it by a calculation rather than a search through states? To make the question sharp we first strip a generation to the object that must be present. Colour plays no active role in what follows: in the model of Part I the strong $SU(3)_c$ is a spectator factor and the quark block is $(\mathbf{3}, R)$ with R a representation of $SU(4)$, so “colour triplet” is automatic and the electroweak content lives entirely in R . The weak $SU(2)_L$ is generated by the first simple root of $SU(4)$, and the hypercharge is a Cartan direction commuting with it. Writing a weight of $SU(4)$ in the partition coordinates (n_1, n_2, n_3, n_4) and the two independent $SU(2)_L$ -invariant charges

$$q_8 = n_1 + n_2 - 2n_3, \quad q_{15} = n_1 + n_2 + n_3 - 3n_4,$$

the unbroken hypercharge that reproduces the Standard-Model quark charges is $Y = (-7q_8 + 4q_{15})/18$ [1], a specific element of the two-parameter family of traceless $SU(2)_L$ -invariant functionals $Y = \alpha q_8 + \beta q_{15}$.

A representation *contains the cell* if its zero modes include a left-handed weak doublet Q and two right-handed weak singlets u, d whose charges under a single such Y are $\frac{1}{6}, \frac{2}{3}, -\frac{1}{3}$. Because Y is linear and homogeneous, this is an exact linear-algebra condition. Collecting the charge vectors of the three multiplets, the cell closes precisely when

$$\det \begin{pmatrix} q_8^Q & q_{15}^Q & \frac{1}{6} \\ q_8^u & q_{15}^u & \frac{2}{3} \\ q_8^d & q_{15}^d & -\frac{1}{3} \end{pmatrix} = 0 \quad \text{with the top } 2 \times 2 \text{ minor nonzero,} \quad (1)$$

i.e. the target vector $(\frac{1}{6}, \frac{2}{3}, -\frac{1}{3})$ lies in the span of the two charge columns, so a unique Y pins all three values. Equation (1) turns “contains the Standard Model” into a statement about the affine geometry of the weight diagram.

One consequence is immediate and organizing. Since $\frac{1}{6} = \frac{1}{2}(\frac{2}{3} + (-\frac{1}{3}))$, the doublet charge is the *mean* of the two singlet charges; if the doublet sits at the midpoint $Q = \frac{1}{2}(u + d)$ of the singlets in charge space, then $Y(Q) = \frac{1}{6}$ holds *automatically* once $Y(u) = \frac{2}{3}$ and $Y(d) = -\frac{1}{3}$. The cell is a symmetric $\pm\frac{1}{2}$ reflection pair straddling the doublet — the geometric face of the half-century-old Pati–Salam hypercharge relation $Y = T_R^3 + (B - L)/2$ [4] (with $Q_{\text{em}} = T^3 + Y$, so $T_R^3 = 0$ for the doublet and $\pm\frac{1}{2}$ for the singlets, and $B - L = \frac{1}{3}$ for a quark); the very $\pm\frac{1}{2}$ that names the cell is Pati and Salam’s right-handed isospin.¹ The whole problem is now: for which

¹A caution against a natural pun: the group carrying our cell is the electroweak $SU(4)$ of the AHMN model, with colour a *separate* $SU(3)_c$; it is not the colour $SU(4)_c$ of Pati and Salam, in which lepton number is a fourth colour. What we borrow from [4] is the hypercharge relation, not the group.

(a, b, c) does the weight diagram of R offer such a configuration among its parity-selected zero modes?

Established here: containing the Standard Model reduces to an exact rank condition (1) on the weight diagram, with the $\pm\frac{1}{2}$ cell a symmetric reflection pair. *Left open:* which labels (a, b, c) actually offer that configuration — the three gates of the rest of the paper.

3 The centre gate (classical)

Before asking *where* the charges sit, a blunter question: which representations can display the Standard-Model fractions $\frac{1}{6}, \frac{2}{3}, \frac{1}{3}$ *at all*? The first obstruction is arithmetic and is not ours. The value of a hypercharge on a weight is, up to normalization, an integer combination of the labels divided by the order of the centre; the Standard-Model fractions $\frac{1}{6}, \frac{2}{3}, \frac{1}{3}$ can be reached only from a fixed congruence class. For $SU(4)$ the relevant invariant is the $\mathbb{Z}_4$ centre charge, equal to the n -ality

$$c(R) = a + 2b + 3c \pmod{4}, \quad (2)$$

Slansky's congruency class [10]. A direct check on the hypercharge above shows the quark fractions occur only when this class is *odd*. That fractional hypercharge is locked to the centre charge in exactly this way is a classical fact, stated for the Standard Model's own $\mathbb{Z}_6 = \mathbb{Z}_2 \times \mathbb{Z}_3$ by Hucks and by Saller, and recast in one-form / global-structure language by Tong and by Alonso, Dimakou and West [13, 14, 15, 16]. We claim none of it; we record that, transferred to the $\mathbb{Z}_4$ centre of the embedding group, it is the first of our three clauses.

Established here: gate one, $(a+2b+3c)$ odd, is the classical centre-charge congruence, inherited not invented. *Left open:* the two geometric gates, which the weight diagram — not the centre — must supply.

4 Delete the middle node: the Levi tower and its chiral projection

The centre says the fractions are reachable. But reachable *together*, under one hypercharge — and where, in $SU(4)$ itself, is that decided? The remaining two clauses, and the reason for them, come from a single structural move. The Dynkin diagram of $SU(4)$ is a path of three nodes, $\circ - \circ - \circ$; *deleting the middle node* splits it into two disjoint $SU(2)$'s and leaves one $U(1)$:

$$SU(4) \supset \underbrace{SU(2)_L}_{\text{node } \alpha_1} \times \underbrace{SU(2)_R}_{\text{node } \alpha_3} \times \underbrace{U(1)_m}_{\text{deleted } \alpha_2}. \quad (3)$$

$SU(2)_L$ is the weak isospin. The deleted node carries the $U(1)$ charge

$$m := 2q_8 + q_{15} = 3(n_1 + n_2 - n_3 - n_4), \quad (4)$$

which — as one checks on the simple roots — is changed only by the middle root α_2 and left fixed by α_1, α_3 : it is the coordinate conjugate to the middle node. This single observation controls both remaining gates.

Branch an irreducible (a, b, c) to this Levi. Its $SU(2)_L$ -*singlet* sector — the seat of the right-handed quark singlets u, d — has a clean shape (Figure 2, verified on all representations to dimension 700): it consists of

$$(b+1) \text{ copies of the Clebsch–Gordan tower } \left[\frac{a}{2}\right] \otimes \left[\frac{c}{2}\right] = \bigoplus_{j=|a-c|/2}^{(a+c)/2} [j], \quad (5)$$

one copy at each of $b+1$ equally spaced values of m (spacing 12, so the singlets span a middle-node extent of exactly $12b$). The number of copies is set by the middle label b ; the internal $SU(2)_R$ spin content is set by the outer labels a, c , with *maximal* spin $(a+c)/2$.

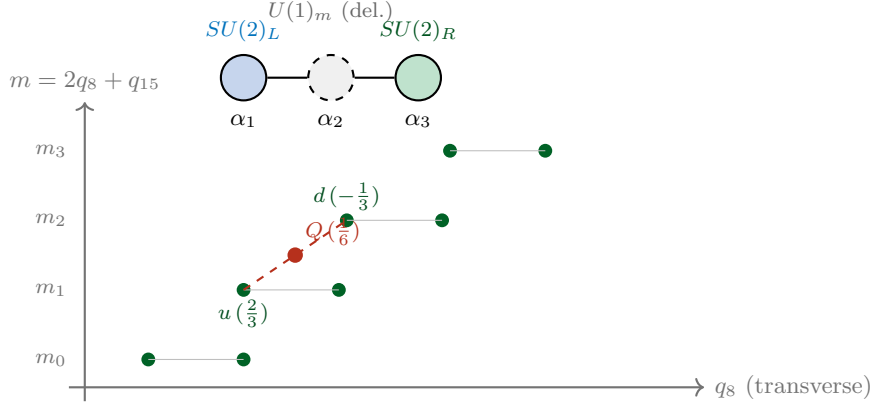


Figure 2: Deleting the middle node of $SU(4)$ exposes $SU(2)_L \times SU(2)_R \times U(1)_m$. The right-handed colour-singlet zero modes (green) form a sheared lattice: $b + 1$ levels in the middle-node charge m (here $b = 3$, four levels), each carrying half of a Clebsch–Gordan tower. A Standard-Model cell (red/green, dashed) is a symmetric $\pm \frac{1}{2}$ pair u, d whose midpoint is a weak doublet Q at hypercharge $\frac{1}{6}$. The cell can close only when the lattice is genuinely two-dimensional — which needs the middle node excited ($b \geq 1$) and at least three singlets ($a + b + c \geq 3$).

Now impose the orbifold. On $T^2/\mathbb{Z}_2$ the two $\mathbb{Z}_2$ parities assign chirality; on the singlet sector they are locked (one is redundant), so the surviving right-handed zero modes are exactly *half* of each tower, and — because the distinct hypercharges are set by the tower’s *extent*, i.e. its top spin $(a + c)/2$ — the count of distinct right-handed colour-singlet charge slots is

$$N(a, b, c) = (b + 1) \cdot \frac{a + c + 1}{2} \quad (6)$$

(the outer factor is an integer because $a + c$ is odd whenever the centre charge is odd). Equation (6) is the technical heart of the paper: a closed-form count of chiral zero-mode multiplets as a function of the bulk representation’s Dynkin labels. The count factorizes as

$$N = \underbrace{(b + 1)(a + c + 1)}_{\text{Levi branching (dimension-independent)}} \times \underbrace{\frac{1}{2}}_{\text{orbifold projection}},$$

a form that makes the roles transparent: the group theory is inherited (a Littlewood–Richardson evaluation [19, 20]), while the halving is the extra-dimensional projection. It is the *extremal chiral projection* — keep the top spin, halve by parity — that is ours, not the closed form it produces.

Established here: a closed chiral zero-mode count $N = (b + 1)(a + c + 1)/2$, with middle-node extent $12b$, from one Levi branching and the parity projection. *Left open:* how this count decides whether the cell closes — the criterion of §5.

5 The cell criterion

We now hold an exact count of the singlets; when is a count enough to close a cell? The count (6) decides the last two gates. The cell (1) needs a *two-dimensional* singlet lattice: three non-collinear charge slots, so that some symmetric pair has a doublet at its midpoint. Reading this off the tower:

- if $b = 0$ there is a single value of m : all singlets are collinear, and no hypercharge can separate the cell. Hence $b \geq 1$ is necessary — the middle node must be excited to open the transverse (spin- $SU(2)_R$) direction; the extent is exactly $12b$.
- with $b \geq 1$, the number of distinct singlets is $N = (b + 1)(a + c + 1)/2$; this drops to $N = 2$ only at the single corner $b = 1$, $a + c = 1$, i.e. $a + b + c = 2$, where the two available singlets

Gate	Inherited (cite)	Ours
$(a+2b+3c)$ odd (<i>arithmetic</i>)	$\mathbb{Z}_4$ centre / n -ality congruence; fractional Y locked to the centre [13, 14, 15, 16, 10]	transfer to the $\mathbb{Z}_4$ of the embedding group as gate 1 of a single closed criterion
$b \geq 1$ (<i>geometric</i>)	Levi branching / Clebsch–Gordan tower [19, 20]	the middle-node extent = $12b$; its necessity for a chiral doublet at $Y = \frac{1}{6}$
$a + b + c \geq 3$ (<i>size</i>)	—	the closed chiral zero-mode count $N = (b+1)(a+c+1)/2$ and its $N \geq 3$ threshold

Table 1: The three gates of Theorem 1, and an honest ledger of what each owes to classical group theory and what is new here. The novelty is the chiral projection giving the closed count, and the packaging of all three into one criterion — not the centre arithmetic nor the branching.

are collinear and the cell is over-constrained. For every other case $N \geq 3$ and the lattice is two-dimensional. Hence $a + b + c \geq 3$.

Together with the centre gate this gives the criterion, which we have verified exhaustively against the direct construction (1) on all representations to dimension 900.

Theorem 1 (Cell criterion). *A finite-dimensional irreducible representation (a, b, c) of $SU(4)$ contains a full Standard-Model quark cell $\{Q(\mathbf{2}, \frac{1}{6}), u(\mathbf{1}, \frac{2}{3}), d(\mathbf{1}, -\frac{1}{3})\}$ among its right/left-handed $T^2/\mathbb{Z}_2$ chiral zero modes if and only if*

$$(a + 2b + 3c) \text{ is odd} \quad \wedge \quad b \geq 1 \quad \wedge \quad a + b + c \geq 3$$

Equivalently: the centre charge is odd (arithmetic), the middle node is excited so the right-handed colour-singlet lattice spreads over $12b$ in the middle-node charge (geometric), and the lattice carries at least three non-collinear points $N = (b + 1)(a + c + 1)/2 \geq 3$ (size).

Theorem 1 is the central result of this paper, and the criterion is ours: the closed three-gate statement, and the chiral-projection count (6) that produces it, are new. We say so without overclaiming — Table 1 records the three clauses and, beside each, exactly what is inherited from classical group theory and what is new, so ownership is explicit gate by gate. The criterion is sparse: up to dimension 400 only eight representations qualify (up to conjugation), of dimensions 60, 84, 140, 140, 216, 224, 280, 360, and the smallest, $(a, b, c) = (0, 2, 1)$ of dimension 60, is precisely the $(3, \mathbf{60})$ of Part I — so Theorem 1 recovers, and explains, the minimality found there by brute force.

Established here: the exact criterion (Theorem 1), which recovers the minimality of Part I as its smallest instance. *Left open:* why $SU(4)$, of all groups, makes the cell a rigid gate at all — the subject of §7.

6 An aside on node-independence

Why is that count so clean — is $(a+1)(b+1)(c+1)$ an accident of $SU(4)$, or does the group insist on it? The clean shape of (6) rests on a small general fact worth isolating, if only to see where $SU(4)$ ’s tidiness ends. For a simple root α , write $\dim \text{Inv}_\alpha(V)$ for the dimension of the subspace of a representation V annihilated by the $SU(2)$ generated by α .

Lemma 1. *For any irreducible representation of a simple Lie algebra, $\dim \text{Inv}_\alpha(V)$ depends only on the Weyl orbit — equivalently the length — of the simple root α . In a simply-laced algebra (all roots one length, hence one Weyl orbit) it is therefore independent of which simple root is chosen; in a non-simply-laced algebra it takes at most two values, one for long and one for short roots.*

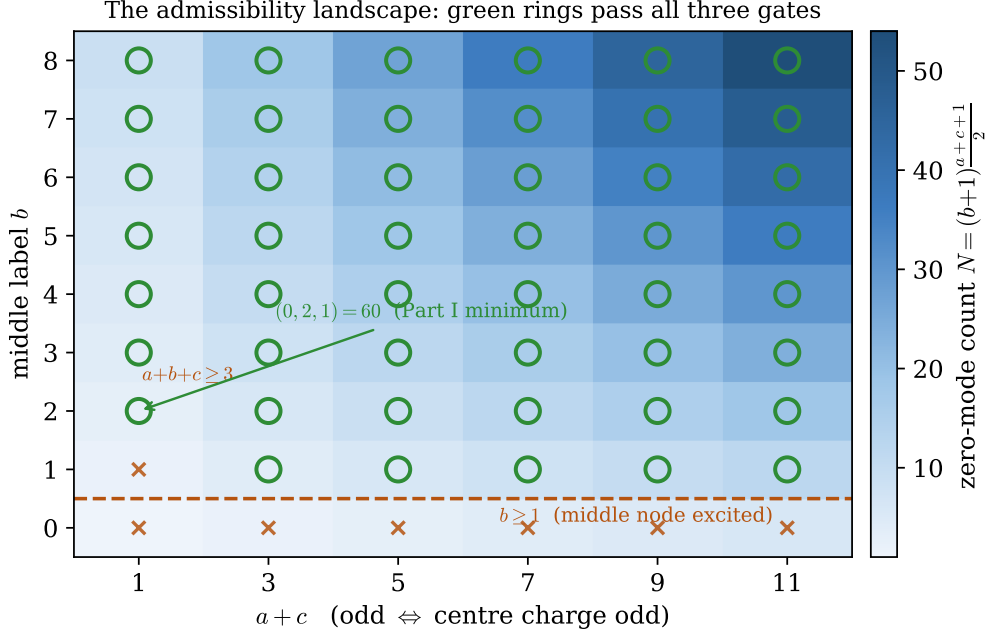


Figure 3: The admissibility landscape of Theorem 1, over the label plane $(a+c, b)$ (the centre gate restricts to $a+c$ odd). The background is the exact zero-mode count $N = (b+1)(a+c+1)/2$; a green ring marks a representation passing all three gates, a red cross one that fails. The gates carve the admissible region cleanly: the whole $b=0$ row falls (middle node not excited), as does the single corner $a+c=1, b=1$ (too small, $N=2$). The minimum, $(0, 2, 1) = 60$ of Part I, sits at the lower-left corner of the admissible region. Every marker is computed, not schematic.

Proof. Weight multiplicities are Weyl-invariant. For w in the Weyl group, $\langle w\mu, (w\alpha)^\vee \rangle = \langle \mu, \alpha^\vee \rangle$, so the distribution of α -eigenvalues over the (Weyl-stable) weight multiset is the same for α and $w\alpha$; hence so is $\dim \text{Inv}_\alpha = N_\alpha(0) - N_\alpha(2)$. In a simply-laced algebra all roots lie in one Weyl orbit. $\square$

For $SU(4)$ (type A_3 , simply-laced) the invariant dimension is thus the same at every node, and equals the clean product $(a+1)(b+1)(c+1)$. This product form, however, is special to A_3 : it is the linear factor of the A_3 Weyl dimension polynomial, and it fails already for $SU(5)$, where the node-independent value is a Littlewood–Richardson sum with no such factorization (for instance it takes the prime value 11 at labels $(1, 1, 0, 0)$, which no product of linear factors reproduces across the lattice). We verified Lemma 1 on types $A_2 - A_4$, D_4 (node-independent) and B_2, B_3, C_3, G_2, F_4 (two values, split exactly by long/short root, e.g. F_4 giving $[21, 21, 15, 15]$). The lemma is elementary and likely folklore; we include it because it identifies precisely which of $SU(4)$ ’s coincidences is generic (node-independence) and which is exceptional (the product), and the latter is one of several senses in which $SU(4)$ is special — the subject of the next section.

7 Why $SU(4)$? The critical rank

Is the criterion a statement about the Standard Model, or about $SU(4)$? Enlarge the group, and the answer separates the two. Theorem 1 is about $SU(4)$; whether the same rigidity survives for larger groups sharpens what “ $SU(4)$ ” is really doing. The cell imposes three charge constraints, $Y(Q) = \frac{1}{6}$, $Y(u) = \frac{2}{3}$, $Y(d) = -\frac{1}{3}$, on a hypercharge functional that, being traceless and $SU(2)_L$ -invariant, has $\text{rank}(G) - 1$ free parameters. The count is the whole story:

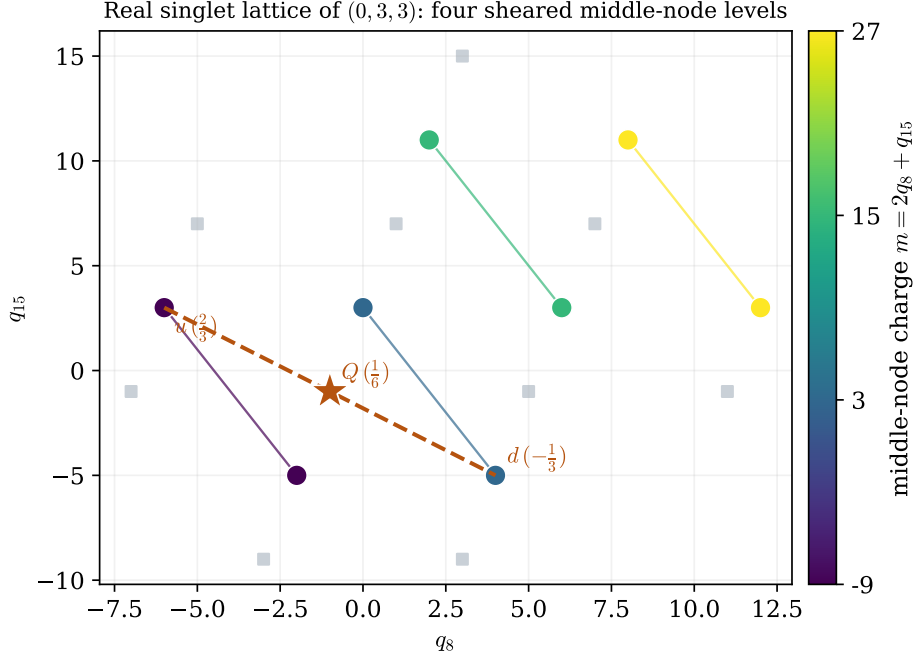


Figure 4: The right-handed colour-singlet weight lattice of a concrete admissible representation, $(0, 3, 3)$, computed exactly. The singlets (filled, coloured by the middle-node charge $m = 2q_8 + q_{15}$) form the sheared $(b+1) \times \frac{a+c+1}{2}$ lattice of §4 — here four m -levels ($b = 3$). Faint grey squares are the weak doublets. A Standard-Model cell (red) is a symmetric $\pm \frac{1}{2}$ pair u, d whose midpoint is a doublet Q ; the shear is exactly what lets three points fail to be collinear, so the cell can close. This is the data behind the schematic of Figure 2.

group	rank	Y parameters	the cell
$SU(4)$	3	2	over-determined — rigid arithmetic gate
$SU(5)$	4	3	marginally solvable — gate dissolves
$SU(6)$	5	4	under-determined — generic reps admit

With three constraints and $\text{rank}(G) - 1$ parameters, the cell is a genuine obstruction (a measure-zero alignment) exactly when $\text{rank}(G) - 1 < 3$, i.e. at rank = 3. Among simple groups large enough to contain the cell, $SU(4)$ is the unique marginal case; at rank 4 and above a *generic* anomaly-free representation admits, the only residual obstruction being too few surviving modes. We have checked this directly: with a free hypercharge, 73% of $SU(5)$ and 85% of $SU(6)$ representations (to dimension 200) admit, across all centre classes, whereas $SU(4)$ stays sparse and centre-selective. The “rank $\geq$ something” arithmetic is classical; what the cell adds is the reading that $SU(4)$ sits exactly at its margin.

There is an honest caveat that is also the useful physics. “Dissolves at higher rank” assumes the hypercharge is *free* to be any surviving combination. In a real model the hypercharge is often *pinned* — fixed, say, by the requirement $\sin^2 \theta_W = \frac{1}{4}$ that motivates these $SU(4)$ constructions. Pinning consumes the free parameters, and then the obstruction can return at any rank. This is exactly what happens in the recent $SU(7)$ grand gauge–Higgs model of Komori and Maru, where the bulk hypercharge generator, fixed to reproduce $\sin^2 \theta_W$, assigns only integer and half-integer charges, “so the fractional hypercharges of quarks cannot be obtained from the [bulk] fields” and the quarks are forced onto a brane [18]. Our counting reads their integer/half-integer fact as a consumed-degree-of-freedom result: with the hypercharge pinned, no residual Cartan freedom remains to place the fractional cell on bulk colour triplets. The robust half of the story is the one that does not depend on the assumption: $SU(4)$ is rigid *even under maximal hypercharge freedom*, and that is why its cell criterion is as sharp as Theorem 1.

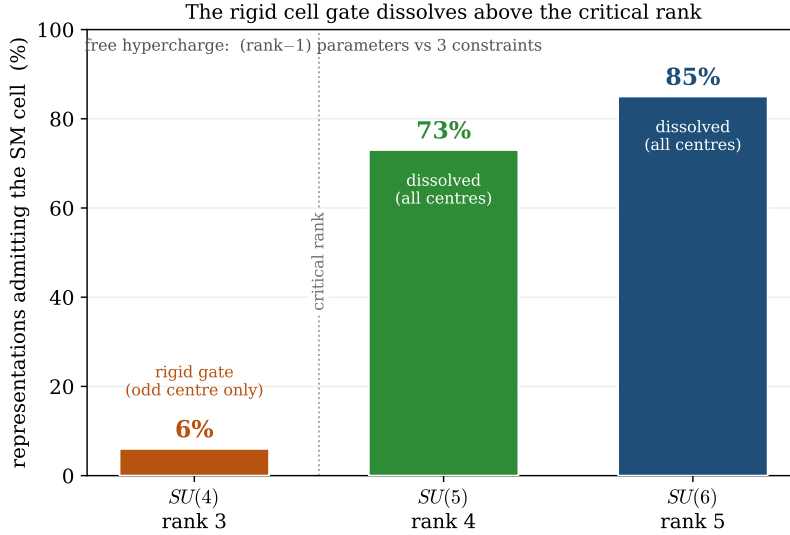


Figure 5: The rigid cell gate dissolves above the critical rank. With a free hypercharge functional ($\text{rank}(G) - 1$ parameters against three constraints), $SU(4)$ (rank 3) is over-determined and admits only a sparse, centre-selective set; $SU(5)$ and $SU(6)$ (rank ≥ 4) become generically solvable, and a large fraction of representations admit across *all* centre classes (measured to dimension 200). The transition is the counting of §7, not a property of any one group.

Established here: $SU(4)$ is the unique rank at which the cell is a rigid obstruction under a free hypercharge, and the same counting explains the higher-rank bulk-quark obstruction as a consumed-freedom effect. *Left open:* the precise $\mathbb{Z}_N$ -orbifold and general- n dependence, laid out next.

8 Reflections and proposals

Three threads seem worth pulling.

The cell is the invariant. Across the groups above, the gauge algebra, the representation, the dimension and the centre all change; the $\pm \frac{1}{2}$ cell does not. It is the weak-isospin skeleton of one generation, the same object in every embedding, and the criterion is best read not as “which representations of $SU(4)$ ” but as “does the group contain the cell in the correct centre sector, with room to close it.” The centre supplies the arithmetic, the geometry supplies the room; the cell supplies the target.

Dimension is a second knob. The count factorizes as (Levi branching) $\times$ (orbifold projection). The first factor is dimension-independent; the second is where the compactification enters. A single $\mathbb{Z}_2$ acting on the chirality axis halves the modes; a $\mathbb{Z}_2$ locked to an existing axis — as the second $T^2/\mathbb{Z}_2$ parity is on the singlet sector, so that five- and six-dimensional counts here coincide — does not; a $\mathbb{Z}_N$ orbifold retains a fraction $1/N$. Higher-dimensional or higher- N projections therefore *shrink* the mode count, so that the geometric-smallness obstruction (the only one left at high gauge rank) can revive. Gauge rank and orbifold projection are opposite knobs on the same cell: one dissolves the arithmetic gate, the other tightens the geometric one. Making the $\mathbb{Z}_N$ dependence a precise counting theorem is a clean next step.

A closed form beyond $SU(4)$. Lemma 1 leaves open the exact node-independent invariant dimension for $SU(n)$: it is a Littlewood–Richardson sum whose $SU(4)$ specialization happens to be a product. Whether it admits a hook-content or determinantal closed form — and whether the

“triangular” $SU(5)$ values $\binom{k+2}{2} = \dim \text{Sym}^k(\mathbb{C}^3)$ generalize to $\dim \text{Sym}^k(\mathbb{C}^{n-2})$ for all symmetric powers — is a well-posed combinatorial question that our data already constrains.

9 Conclusion

We have turned the minimality observed in Part I into an exact rule. An $SU(4)$ representation (a, b, c) contains a Standard-Model quark cell on $T^2/\mathbb{Z}_2$ if and only if its centre charge is odd, its middle node is excited, and its labels sum to at least three; the derivation runs through a Levi branching whose Clebsch–Gordan tower, projected onto the orbifold’s chiral zero modes, yields the closed count $N = (b+1)(a+c+1)/2$. We have been deliberate about credit: the centre arithmetic, the branching, and the hypercharge-parameter counting are classical, and we cite them as such; what is ours is the chiral projection that produces the closed count, the packaging into a single criterion, and the reading of the $\pm\frac{1}{2}$ cell as the invariant that places $SU(4)$ at the critical rank. The last of these also explains, rather than merely restates, why higher-rank constructions with a pinned hypercharge cannot keep their quarks in the bulk. The criterion is exact and machine-checked; the questions it opens — the general- n closed form and the $\mathbb{Z}_N$ dimension dependence — are stated precisely enough to be settled. And it closes back onto the thread Part I left hanging: viability is decided model by model, but a model must first be *eligible*, and Theorem 1 is the eligibility test — it says which representations can even be entered into the search for a viable six-dimensional $SU(4)$ generation. From one witness to the landscape it lives in, that is the arc of the two papers.

Reproducibility. Every count regenerates from the ancillary SageMath scripts: the direct cell construction (1), the Levi-branching derivation of (6), the exhaustive criterion check to dimension 900, the $SU(5)/SU(6)$ rank scans, and the node-independence sweep across Lie types of Lemma 1.

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